

Fundamentals in Computational Analysis of Large-Scale Datasets

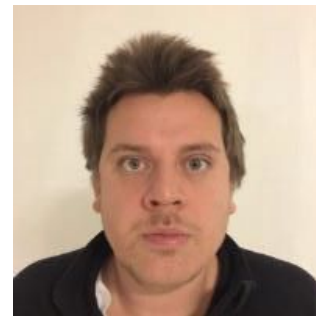
7 September 2026

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About me

- Bachelor Mathematics and computer-science. 2008
- Master Bioinformatics. 2012
- PhD at NHM. 2015
- PostDoc at Cambridge, UK. 2017
- Associate Professor NHM. 2018
- *Section for GeoGenetics* at the GLOBE institute



I implement and develop method for the analyses of aDNA and low-coverage sequencing data.



Overview: Day 1-UNIX shell (7 September)

- **Back to basics**
- UNIX and POSIX
- Command line interface (CLI),
shell
- Filesystem, Files, Directories, paths
- Programs, processes, job control
- stdin/stdout, pipes, filters
- Loops, scripts
- Transferring files between systems

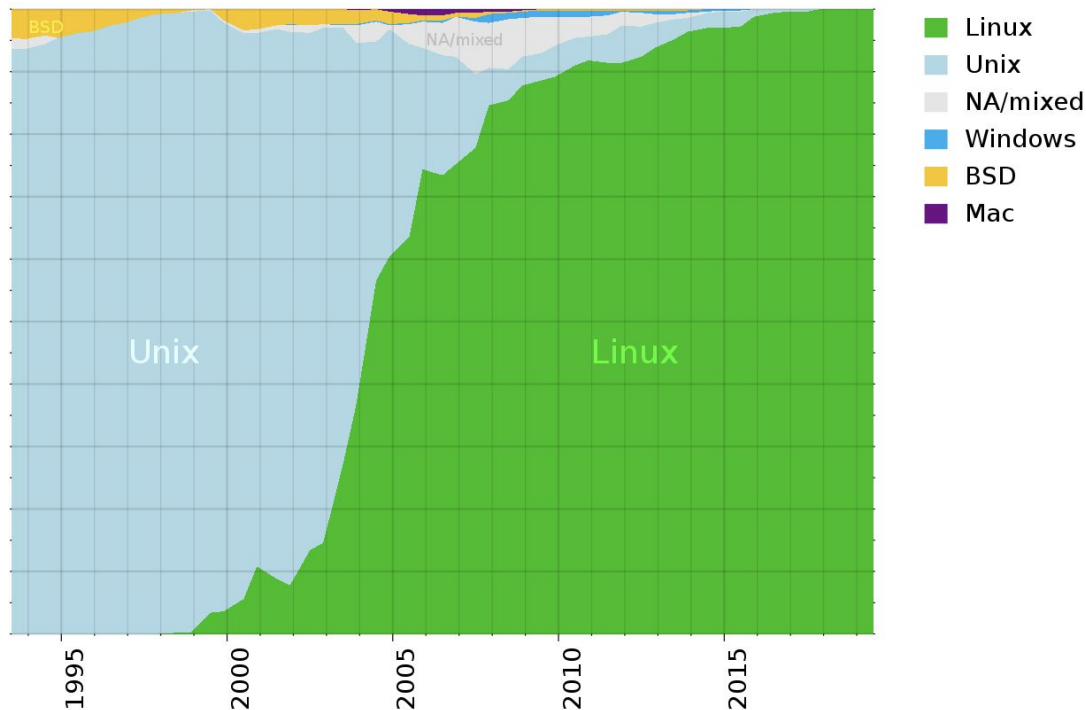
```
[root@localhost ~]# ping -q fa.wikipedia.org
PING text.pmtpa.wikimedia.org (208.80.152.2) 56(84) bytes of data.
^C
--- text.pmtpa.wikimedia.org ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 540.528/540.528/540.528/0.000 ms
[root@localhost ~]# pwd
/root
[root@localhost ~]# cd /var
[root@localhost var]# ls -la
total 72
drwxr-xr-x. 18 root root 4096 Jul 30 22:43 .
drwxr-xr-x. 23 root root 4096 Sep 14 20:42 ..
drwxr-xr-x.  2 root root 4096 May 14 00:15 account
drwxr-xr-x. 11 root root 4096 Jul 31 22:26 cache
drwxr-xr-x.  3 root root 4096 May 18 16:03 db
drwxr-xr-x.  3 root root 4096 May 18 16:03 empty
drwxr-xr-x.  2 root root 4096 May 18 16:03 games
drwxrwx--T.  2 root gdm  4096 Jun  2 18:39 gdm
drwxr-xr-x. 38 root root 4096 May 18 16:03 lib
drwxr-xr-x.  2 root root 4096 May 18 16:03 local
lrwxrwxrwx.  1 root root    11 May 14 00:12 lock -> ../run/lock
drwxr-xr-x. 14 root root 4096 Sep 14 20:42 log
lrwxrwxrwx.  1 root root    10 Jul 30 22:43 mail -> spool/mail
drwxr-xr-x.  2 root root 4096 May 18 16:03 nis
drwxr-xr-x.  2 root root 4096 May 18 16:03 opt
drwxr-xr-x.  2 root root 4096 May 18 16:03 preserve
drwxr-xr-x.  2 root root 4096 Jul  1 22:11 report
lrwxrwxrwx.  1 root root    6 May 14 00:12 run -> ../run
drwxr-xr-x. 14 root root 4096 May 18 16:03 spool
drwxrwxrwt.  4 root root 4096 Sep 12 23:50 tmp
drwxr-xr-x.  2 root root 4096 May 18 16:03 yp
[root@localhost var]# yum search wiki
Loaded plugins: langpacks, presto, refresh-packagekit, remove-with-leaves
rpmfusion-free-updates                               2.7 kB    00:00
rpmfusion-free-updates/primary_db                    206 kB    00:04
rpmfusion-nonfree-updates                             2.7 kB    00:00
updates/metalink                                     5.9 kB    00:00
updates                                               4.7 kB    00:00
updates/primary_db                                  73% [=====] 62 kB/s | 2.6 MB    00:15 ETA
```

Main interface for accessing servers. Running programs on servers. CLI is also called the shell.



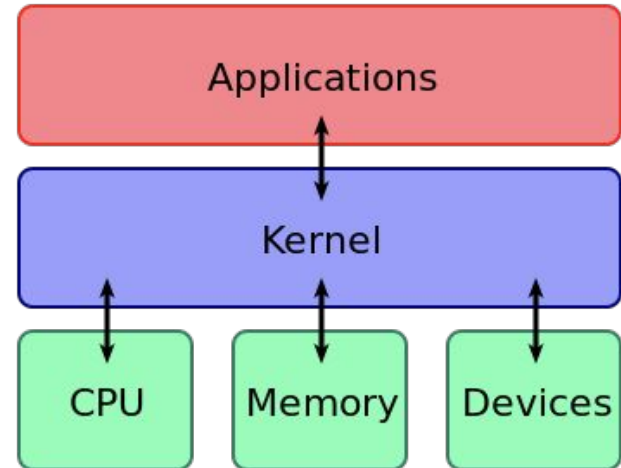
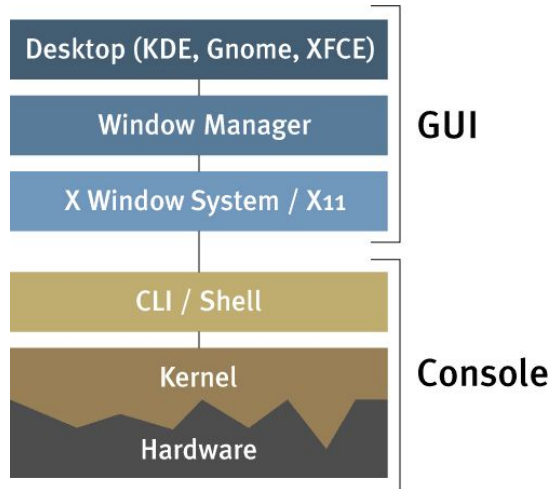
Operating systems: Windows, macOS, UNIX, and Linux

- Most desktop computers are running windows: 77%-87.8%
- All top500 supercomputers in the world is running Linux. (see Figure to the right)
- UNIX systems are everywhere: smartphones, servers, apple laptops, fridges.
- UNIX is access through a shell rather than graphical user interfaces (GUI)
- Shells are used for automating menial tasks
- Shell knowledge is essential for data analysis



Basic concepts

- Hardware/Software: Hardware is the physical part, software are the programs or instructions that 'run' on the hardware
- Focus will be programs and software that are started by 'users' and will run on the linux server(the kernel)



Operating system: Low level software that controls the hardware, scheduling of processes, access to network, screen, mouse etc.

Our operating system is UNIX (follows POSIX standard) which means it has to have a shell with a number of tools supplied by the operating system.



- Begins with prompt. Server is now ready for input.
- Type command press <enter>
- The command is executed
- A new prompt is displayed ready for next input
- Sometimes the prompt starts with \$ instead of >

(When a program is executed it is a process)



Example of shell commands

```
testfolder — -bash — 80x24
[thorfinns-imac:testfolder thorfinn$ ls
myscript.sh      subdirectory      subfolder          testfile.txt
[thorfinns-imac:testfolder thorfinn$ ls -F
myscript.sh*     subdirectory/     subfolder/         testfile.txt
[thorfinns-imac:testfolder thorfinn$ date
Sun Mar  7 15:19:17 CET 2021
[thorfinns-imac:testfolder thorfinn$ whoami
thorfinn
[thorfinns-imac:testfolder thorfinn$ echo loremipsum >subdirectory/file.txt
[thorfinns-imac:testfolder thorfinn$ ls -FR
myscript.sh*     subdirectory/     subfolder/         testfile.txt

./subdirectory:
file.txt

./subfolder:
thorfinns-imac:testfolder thorfinn$
```



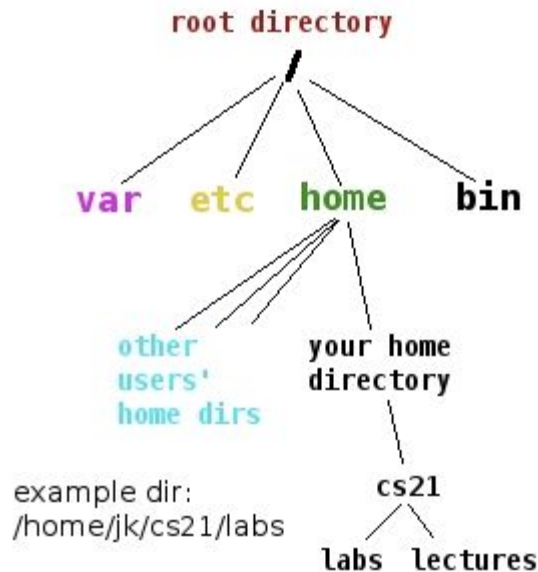
UNIX Files and filesystem 1/3

- **Computer file:** Is a sequence of bytes that resides on some storage medium.
- **Pathname:** Is a name that uniquely identifies a computerfile in a filesystem.
- **Directory:** Is a container for a group of files. A directory can can be included in other directories.
- For example **dir1/hello**, refers to the file called hello that resides in folder/directory dir1
- For example **/dir1/dir2/dir3/README.txt**. /dir1/dir2/dir3 is the dirname, basename is README.txt it has the **extension** txt.
- Please note the difference between **absolute path** and **relative path**.
- When you log on to a UNIX system you are put into your **home** directory **\$HOME**.



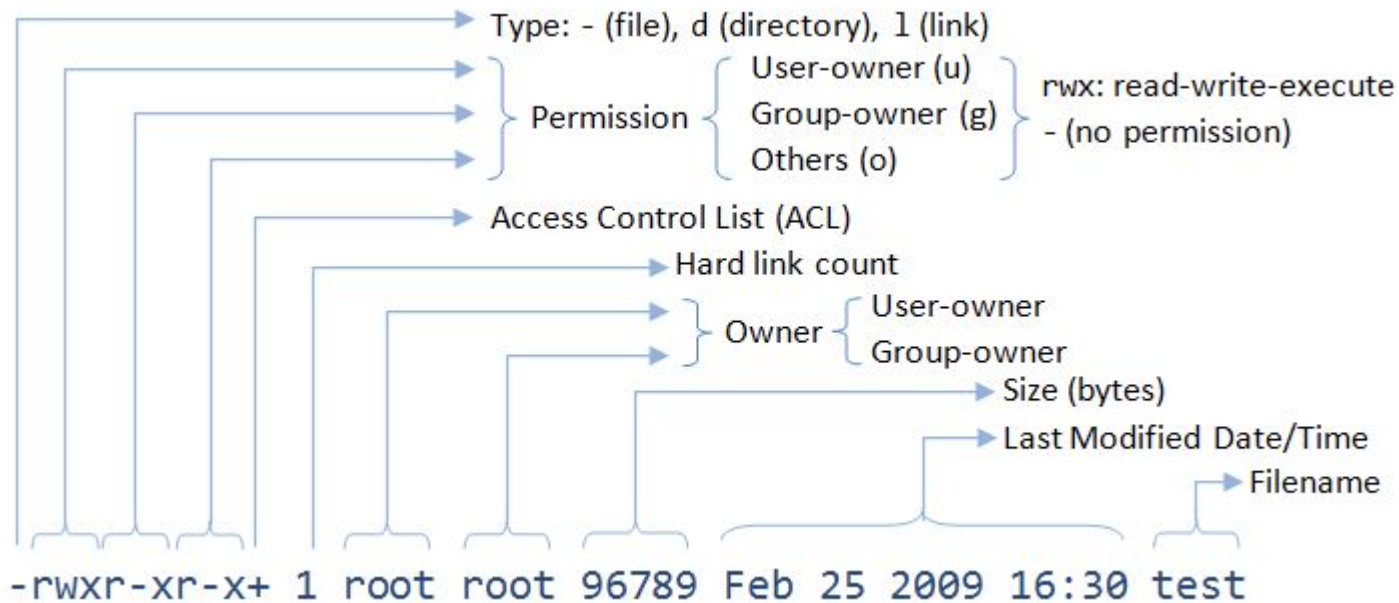
UNIX Files and filesystem 2/3

- List files and directories, **ls**
- Remove/delete a file, **rm**
- Make or create a directory **mkdir**. Delete a directory **rmdir**
- Change to directory, **cd** (change directory)
- Step one level up towards the root, **cd ..**, step two steps up, **cd ../../**
- Get current directory, **pwd**
- Copy a file, **cp** source destination
- Copy a directory (with contents, called recursive), **cp -r** sourcedir destinationdir
- Move a file, **mv** source destination
- Print contents of file **cat** or **tac** or interactively **less**
- Count the number of words, and lines **wc** (wordcount)



UNIX Files and filesystem 3/3

>ls -l test



PRE Exercise 1: Software carpentry (shell)

<https://swcarpentry.github.io/shell-novice/>

- Download the data
- Install the shell



Exercise 1: Software carpentry (shell)

If you have never heard about these concepts before, none of it will have made sense until you have tried it yourself.

THE SETUP:

You are ‘Nelle Nemo’, a marine biologist. Your supervisor has given you a great project: but you have to use *her* analysis tools, and they are command line tools that only work on Unix machines...

GO HERE: <https://swcarpentry.github.io/shell-novice/>

1. [Introducing the Shell](#)
2. [Navigating Files and Directories](#)
3. [Working With Files and Directories](#)

Boxes in orange will indicate questions that you should not run but try to resolve by thinking

Blue boxes have commands you should try to run. The output is shown in grey.



Questions:

1. How many files do we have in the 'data-shell' folder ?
2. How many directories do we have in the 'data-shell' folder ?
3. What does the -r or -R parameter do for most commands 'mv' 'cp' etc?
4. What does the -p option do for the mkdir command ?

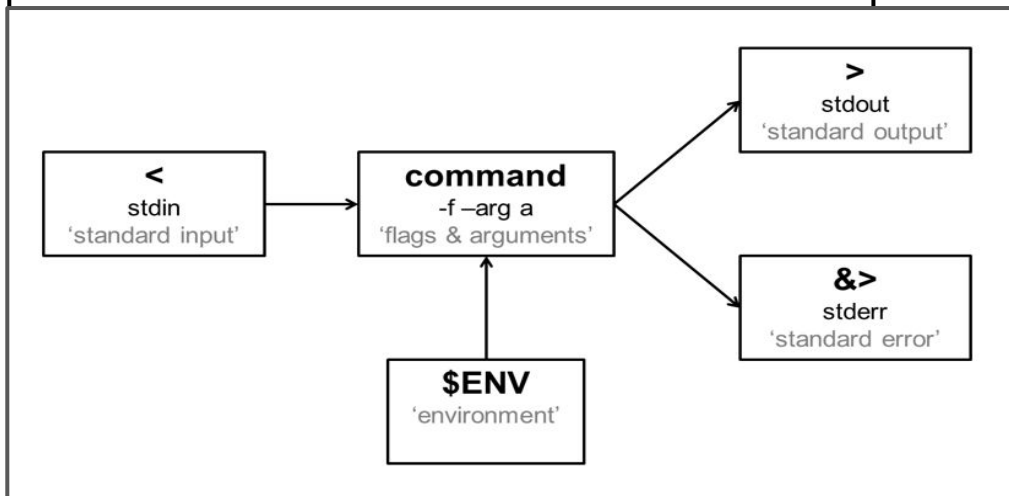


STDIN, STDOUT, STDERR 1/2

Unix processes have some standard ways of handling input and output.

The “environment” is the list of properties the process picks up from its parent. Your processes will all have the shell as their parent.

```
$ tr -d "\t"
$ tr -d "\t" <in >out 2>err
```



STDIN, STDOUT, STDERR 2/2



```
>demoprogram <enter>
                                Hello whats your name?
John <enter>
                                Hello 'John', how old?
Doe <enter>
                                Error 'Doe' is not a number, how old?
54 <enter>
                                Hello John you are 54 years old
>echo $? ##this is returncode
```

- Blue is stdout
- Green is stdin
- Red is stderr

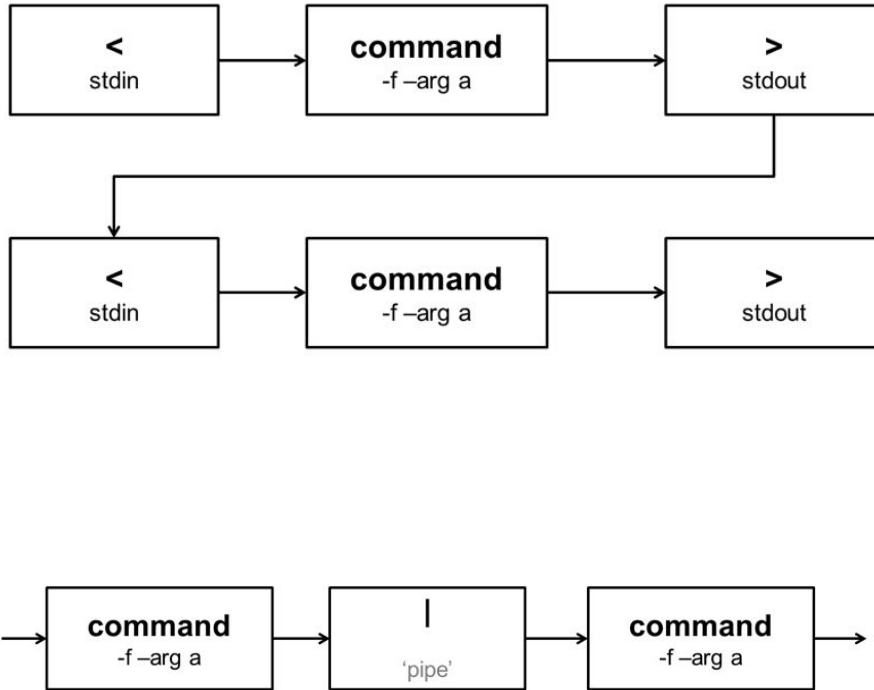


STDIN, STDOUT can be chained (PIPES)



- Example. Count the number of files in a directory.
- `ls | wc <enter>`
ls is program1
wc is program2, wordcount
- The vertical line 'l' (not small letter 'l') is called pipe. It glues stdout to stdin
- Output from a process can be written into a file with '>filename'





The 'tr' program is a command that 'translates' characters.

```
$ echo "abcd" >txt.txt
```

```
$ cat txt.txt
```

```
abcd
```

```
$ tr "d" "e" <txt.txt
```

```
abce
```

```
$ tr "d" "e" <txt.txt >output.txt
```

```
$ cat output.txt
```

```
abce
```

```
$ rm output.txt
```

```
$ tr "d" "e" >output.txt <txt.txt
```

```
$ cat output.txt
```

```
abce
```

```
$ cat txt.txt |tr "d" "e"
```

```
abce
```

Writing scripts

```
#!/bin/bash
```

```
Command1
```

```
Command2
```

```
Command3
```

```
command4
```

#! Is called *shebang* and instructs the shell which program should be used for interpreting the file.

Files containing scripts has to be regular boring textfiles. Not .docx or similar.

sh: bourne shell

csh: C shell

tcsh: TC shell

ksh: Korn shell

bash: bourne again shell

dash: Debian almquist shell

zshell: Default on macOS



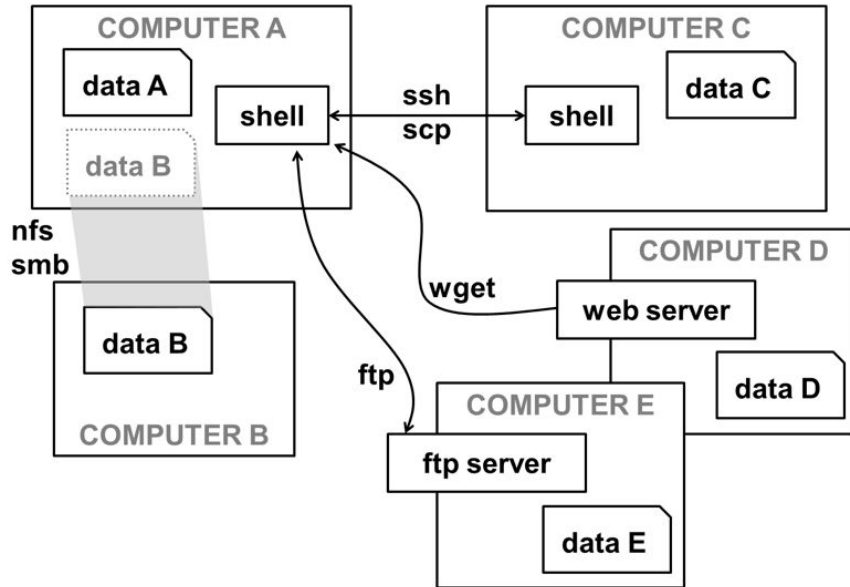
Shell and job control

- Programs can be either terminal programs or graphical programs
- Programs that are started will take focus from terminal until finished, paused (suspended) or put in background
- List of current running process can be found with 'ps'
- CTRL+C kill program
- CTRL+Z suspend program
- bg=background, fg=foreground
- Programs can be started directly in background by adding '&' after program name



Remote systems?

Most of the ways of moving data around the internet were developed for Unix first. You also have the option of going to where the data is, with a remote shell.



ssh: 'secure shell'

Opens up a shell session on a remote machine, over an encrypted channel.

scp: 'secure copy'

Carries out a copy between two machines, using the ssh machinery.

Wget 'web get'

Lets you grab a file using a url, without all that messing around with web browsers.

Ftp: 'file transfer protocol'

Creates another shell-like environment (with a different command set), from which you can connect to other machines and push or retrieve files.

Exercise 2: Software carpentry (shell)

If you have never heard about these concepts before, none of it will have made sense until you have tried it yourself.

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THE SETUP:

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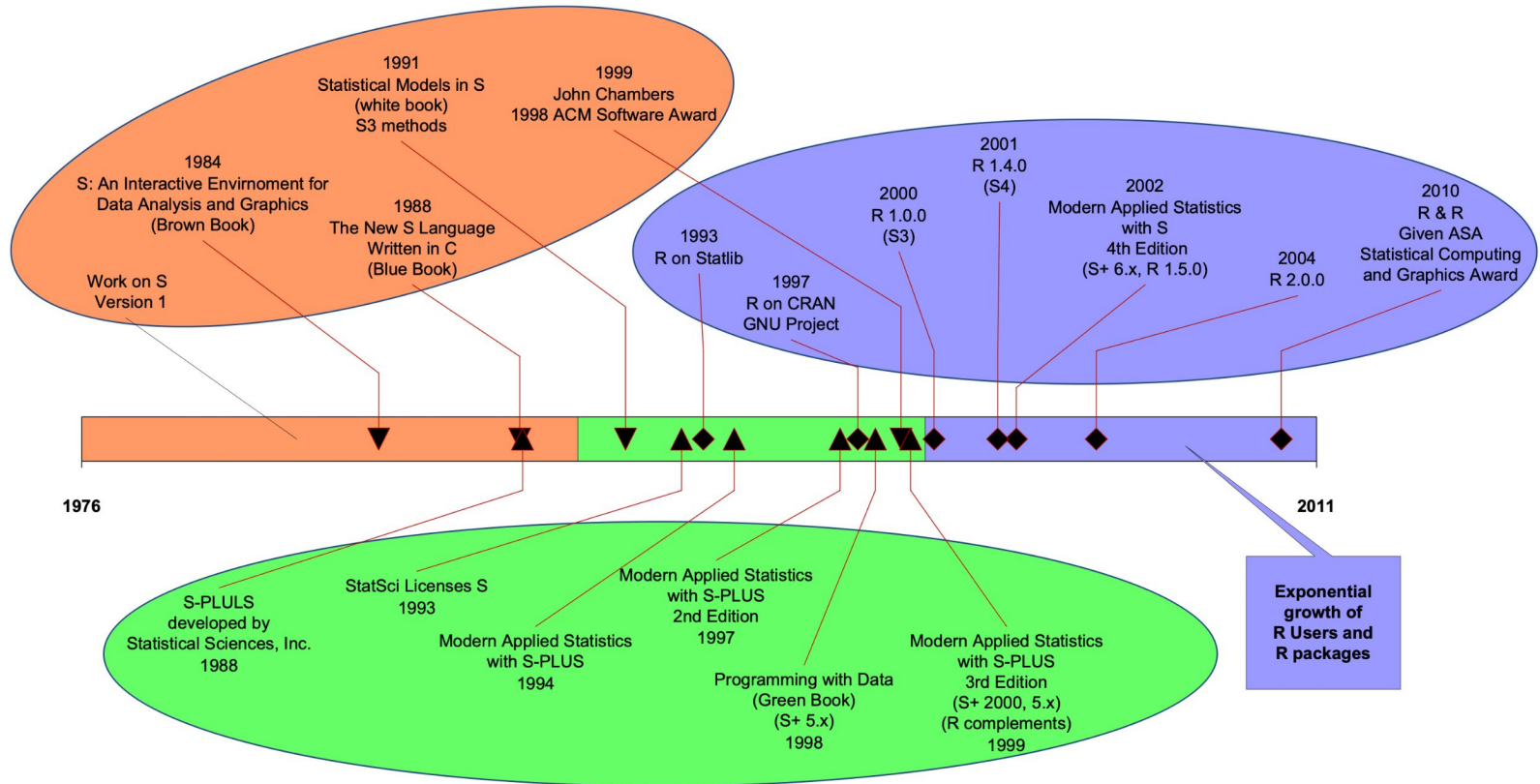
4. Pipes and Filters

5. Loops

6. Shell Scripts



R programming language



R programming language

- Statistical modelling
- Data analysis
- Data manipulation
- Data visualization
- 9th most popular programming language 2021
- R is an interpreted language:
 - When you enter expressions into the R console or R script
 - A program within the R system, called the interpreter, executes the actual code that you wrote



Rstudio is an integrated developer environment (IDE)

The screenshot displays the RStudio integrated development environment (IDE) interface. The top toolbar includes icons for file operations and a search bar. The main window is divided into four panes:

- Console:** Shows the R version (3.6.2), copyright information, and the results of the command `1:10`, which is `[1] 1 2 3 4 5 6 7 8 9 10`. The command `plot(1:10)` has been executed, creating a plot.
- Environment:** Displays the global environment with a single variable `a` of type `int [1:10]`, containing the values `1 2 3 4 5 6 7 8 9 10`.
- Plots:** Shows a scatter plot of the values of `a` against the index. The x-axis is labeled "Index" and ranges from 1 to 10. The y-axis is labeled "a" and ranges from 1 to 10. The plot shows a linear relationship with open circles at each integer value.
- Files:** Shows the file explorer with a "Publish" button.

The bottom status bar indicates the current project is "(None)".

Key Points for September 6

1. Calculator
2. Variables
3. Comments
4. Data types
5. Data structures
6. Builtin functions
7. Writing your own functions
8. Installing packages
9. Simple plotting

Calculator

> 2+3+4+5

[1] 14

- Red text is what you type in
- Blue text is the response from R

Intended Learning Objectives:

- To be able to open and close R
- To be able to use R as calculator



Variables and comments

1. A variable is a placeholder used to represent a specific value.
2. A comment in R starts with `#`
3. Arrowsyntax `<-` General
4. Equalsyntax `=` used in 'function definitions'
5. `Assign()` Notice that here the variable name is allowed to be a variable

```
## Calculate area of circle with radius 10  
> 3.141592653589793*10*10  
[1] 314.1593  
> pi <- 3.141592653589793  
> r <- 10  
> pi*r^2  
[1] 314.1593
```

R Code: Variable assignment

```
> y <- 5  
> y  
[1] 5  
  
> assign("e",2.7183)  
> e  
[1] 2.7183  
  
> s = sqrt(2)  
> s  
[1] 1.414214  
  
> r <- rnorm(n=2)  
> r  
[1] 0.4296685 0.4046568
```



R vectors 1/2

Vectors can be constructed with the 'c()' combine function. Used for most operations that work on scalars.

R Code: Creating vectors and vector operations

```
> constants <- c(3.1416,2.7183,1.4142,1.6180)
> names(constants) <- c("pi","euler","sqrt2","golden")
> constants

      pi  euler  sqrt2  golden
3.1416 2.7183 1.4142 1.6180

> constants^2

      pi      euler      sqrt2      golden
9.869651 7.389155 1.999962 2.617924

> 10*constants

      pi  euler  sqrt2  golden
31.416 27.183 14.142 16.180
```



R vectors 2/2

R Code: Indexing vectors

```
> constants[c(1,3,4)]

    pi  sqrt2 golden
3.1416 1.4142 1.6180

> constants[c(-1,-2)]

    sqrt2 golden
1.4142 1.6180

> constants[c("pi","golden")]

    pi golden
3.1416 1.6180

> constants > 2

    pi  euler  sqrt2 golden
TRUE   TRUE  FALSE  FALSE

> constants[constants > 2]

    pi  euler
3.1416 2.7183
```

```
> constants
```

```
    pi  euler  sqrt2 golden
3.1416 2.7183 1.4142 1.6180
```

Vectors are accessed with the bracket notation `[]`
Examples:

1. Positive integer index ranges (extracting)
2. Negative integer index ranges (discarding)
3. By names
4. By logical vector



Datatypes 1/2

1. Everything in R is an object and all objects have classes
2. Use `ls()` or `objects()` functions to obtain a character vector of the names of the current existing objects
3. Use the `class()` function to obtain the class of the specified object
4. A *datatype* is an attribute of some data that tells the interpreter what the type is.
5. R has six *atomic* datatypes (these can be adjoint to a *vector*)

typeof	mode	storage.mode
logical	logical	logical
integer	numeric	integer
double	numeric	double
complex	complex	complex
character	character	character
raw	raw	raw

Classical classes

- Numeric
- Character
- Data.frame
- Matrix



Datatypes 2/2

```
> x <- c(3.1416,2.7183)
> m <- matrix(rnorm(9),nrow=3)
> tab <- data.frame(store=c("downtown","eastside","airport"),sales=c(32,17,24))
> cities <- c("Seattle","Portland","San Francisco")
```

```
> ls()
[1] "cities" "m"      "tab"    "x"
> objects()
[1] "cities" "m"      "tab"    "x"
> class(m)
[1] "matrix"
> class(x)
[1] "numeric"
```

R Code: Object type (storage mode)

```
> x
[1] 3.1416 2.7183
> typeof(x)
[1] "double"
> cities
[1] "Seattle"      "Portland"
[3] "San Francisco"
> typeof(cities)
[1] "character"
```



Data.frame vs Matrix

Matrix:

- Two-dimensional - $m \times n$
- Similar datatypes (homogeneous)
- Fixed number of rows and columns

Data.frame:

- Generalized form of matrix
- Different data types (heterogeneous)
- Variable number of rows and columns

R Code: Object **class**

```
> m
```

```
          [,1]      [,2]      [,3]  
[1,] -0.6147361 -0.2248133 0.1354078  
[2,] -0.7835507  2.3798959 0.8825350  
[3,]  1.0156090  1.4605885 0.9470563
```

```
> class(m)
```

```
[1] "matrix"
```

```
> tab
```

```
      store sales  
1 downtown    32  
2 eastside    17  
3  airport    24
```

```
> class(tab)
```

```
[1] "data.frame"
```



Data.frame - useful functions

```
> df<-read.table(file="file.csv", sep=";", skip=1)
```

```
> write.table(df,"dfSaved.txt",sep="\t",row.names=FALSE,col.names=TRUE)
```

Slice and Subset of dataframes

row1 is `df[1,]` row2 is `df[2,]`

column1 is `df[,1]` column2 is `df[,2]`

Rename dataframes

```
colnames(df) <- c("Long", "Lat", "City")
```

Bind dataframes

```
rbind(df[,1], df[,2])
```

```
cbind(df[,2], df[,1], df[,1])
```

```
cbind(df$Long, df$City)
```

```
#longitude,latitude,city name
145.768,-16.915,"Cairns"
146.801,-19.265,"Townsville"
150.501,-23.365,"Rockhampton"
139.485,-20.715,"Mount Isa"
150.893,-34.423,"Wollongong"
151.785,-32.932,"Newcastle"
141.451,-31.965,"Broken Hill"
145.951,-30.082,"Bourke"
150.932,-31.091,"Tamworth"
149.581,-33.417,"Bathurst"
153.118,-30.315,"Coffs Harbour"
146.901,-36.065,"Albury"
```



Functions (builtin)

We can sample numbers from a normal (gaussian) distribution with the `rnorm` function.

```
> args(rnorm)
function (x, mean = 0, sd = 1)
NULL
```

```
> rnorm(2)
[1] -0.05526951 1.03702853
> rnorm(2,sd=10)
[1] -10.487585 3.841609
> rnorm(2,sd=10,mean=100)
[1] 108.5422 126.7425
```

- This function takes 3 arguments.
- 2 of the arguments has default values

1. Unnamed arguments are assigned according to position
2. Named argument are assigned by name (partial matching)
3. Only arguments with no default value are required to be supplied



Writing your own functions

Its very easy to write your own functions

```
> myadder <- function(a,b=2){  
  mysum <- a+b  
  return(mysum)  
}  
> myadder(2)  
[1] 4  
>myadder(2,b=3)  
[1] 5
```

- This function takes 2 arguments.
- Second argument is not mandatory

The arguments for a function in R can in itself be a function.

Variables that are defined *within* a function does not exist after the function has run.

You can define functions *within* a function, then these will not be visible outside the function.



Installing packages

- All functions in R are stored in packages
- Most useful functions are included in core packages which is included in a standard R installation

You can develop your own R scripts that parse and analyse some data. But sometimes you need some functionality that is not included in a standard installation of R.

R Code: The `install.packages` function

```
> args(install.packages)

function (pkgs, lib, repos = getOption("repos"), contriburl = contrib.url(repos,
  type), method, available = NULL, destdir = NULL, dependencies = NA,
  type = getOption("pkgType"), configure.args = getOption("configure.args"),
  configure.vars = getOption("configure.vars"), clean = FALSE,
  Ncpus = getOption("Ncpus", 1L), libs_only = FALSE, INSTALL_opts,
  ...)
NULL

> #install.packages("nutshell")
> # or if repos needs to be specified
> #install.packages("nutshell", repos="http://cran.fhcrc.org")
```

After you have installed the package you will need to "load" it

```
> library("nutshell")
```



Plotting data 1/2

- R is very useful as a calculator
- R is very useful for data exploration and manipulation
- R is very useful for visualizing data and making publication ready figures

Below are some of the most important plotting functions

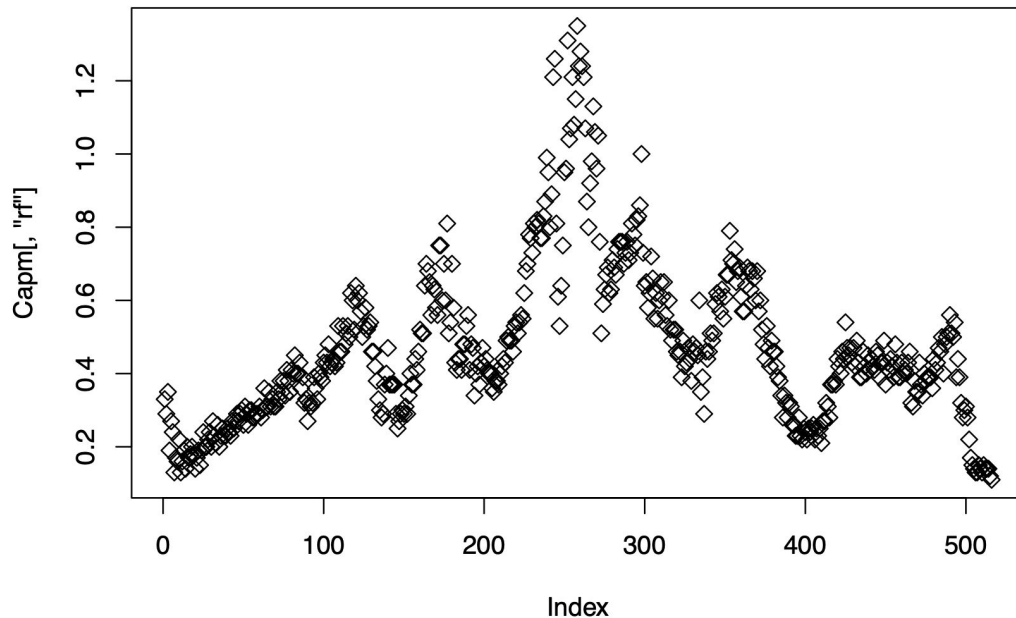
Function	Description
<code>plot</code>	generic function to <code>plot</code> an R object
<code>lines</code>	adds lines to the current <code>plot</code>
<code>segments</code>	adds lines line segments between point pairs
<code>points</code>	adds points to the current <code>plot</code>
<code>text</code>	adds text to the current <code>plot</code>
<code>abline</code>	adds straight lines to the current <code>plot</code>
<code>curve</code>	<code>plot</code> a function over a range
<code>legend</code>	adds a legend to the current <code>plot</code>
<code>matplot</code>	<code>plot</code> all columns of a matrix
<code>par</code>	sets graphics parameters



Plotting data 2/2

R Code: **Plot** with defaults

```
> library(Ecdat)  
> data(Capm)  
> plot(Capm[, "rf"], pch=5)
```



Important topics that has not been covered

1. The data structure called **list()**
2. Writing data to files **write.table()**
3. Factor levels
4. Writing figures to files **pdf(),png()** etc

Some of the examples are based on the following slideshow:

<https://faculty.washington.edu/ezivot/econ424/RIntro.pdf>

